

IIT Kharagpur

Department of Mathematics

Semester-4 (DC)

Statistical Inference

MA41108 (LTP- 3-0-0,CRD- 3, 40 hrs)

Prerequisites :

Probability and Statistics (MA20205)

Parametric point estimation:

- (1) Sampling distributions (revision). [3 Lectures]
- (2) Properties of estimator : Unbiasedness, consistency, efficiency, invariance.[3 Lectures]
- (3) Methods of Estimation: Method of moments and maximum likelihood.[3 Lectures]
- (4) Lower bounds of variance: Frechet-Rao-Cramer, Bhattacharya, Chapman-Robbins-Kiefer lower bounds. [3 Lectures]
- (5) Data reduction : Sufficiency, Factorization Theorem, Fisher's Information, minimal sufficiency, ancillary statistic, completeness and bounded completeness, Basu's Theorem. [5 Lectures]
- (6) Improving estimators: Rao-Blackwell Theorem, Lehmann-Scheffe Theorem,UMVUE [4 Lectures]

Test of hypothesis:

- (7) Hypothesis: Simple and composite hypotheses, types of error. [2 Lectures]
- (8) Construction & comparison of tests: Neyman-Pearson Lemma, UMP, UMP unbiased and UMP invariant tests. [6 Lectures]
- (9) Likelihood ratio tests: applications to one sample and two sample problems, Chi-square tests. Sequential probability ratio test, Families with monotone likelihood ratio. [6 Lectures]

Confidence interval estimation:

- (10) Interval estimation: Methods for finding confidence intervals, shortest length confidence intervals. Bootstrap method for confidence interval. [6 Lectures]

References:

- (1) Casella, G., & Berger, R. L. (2002). Statistical inference (Vol. 2). Pacific Grove, CA: Duxbury. [Text]
- (2) Wasserman, L. (2013). All of statistics: a concise course in statistical inference. Springer Science & Business Media. [Text]
- (3) Kale, B. K ., (2005) A First Course on Parametric Inference.
- (4) Miller, I., Miller, M., & Freund, J. E. (2014). John E. Freund's mathematical statistics with applications. Boston: Pearson,.
- (5) Lehmann, Erich L., and George Casella. (2006) Theory of point estimation. Springer Science & Business Media.
- (6) Lehmann, E. L., & Romano, J. P. (2006). Testing statistical hypotheses. Springer Science & Business Media.
- (7) Bickel, P. J., & Doksum, K. A. (2015). Mathematical statistics: basic ideas and selected topics, volume I (Vol. 117). CRC Press.

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